

Trigonometry & Polar Coordinates



Law of Sines

- 1 In a triangle, $\angle A = 40^\circ$, $a = 10$, and $\angle B = 65^\circ$. Find b , rounded to the nearest hundredth.
 - 2 In a triangle, $\angle A = 30^\circ$, $\angle B = 75^\circ$, and $a = 6$. Find $\angle C$ first, then explain how you would find side c .
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Law of Cosines

- 3 A triangle has $a = 10$, $b = 6$, and included angle $C = 40^\circ$. Find c , rounded to the nearest hundredth.
 - 4 Explain when you should use the Law of Cosines instead of the Law of Sines to solve a triangle.
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Double-angle identities

- 5 If $\sin x = \frac{3}{5}$ and x is in Quadrant I (so $\cos x = \frac{4}{5}$), find $\sin 2x$ and $\cos 2x$.
 - 6 If $\cos x = -\frac{1}{2}$ and x is in Quadrant II (so $\sin x = \frac{\sqrt{3}}{2}$), find $\cos 2x$.
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Polar to rectangular conversion

- 7 Convert the polar point $(8, 150^\circ)$ to rectangular coordinates.
 - 8 Convert the polar point $\left(6, \frac{\pi}{3}\right)$ to rectangular coordinates.
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Rectangular to polar conversion

- 9 Convert the rectangular point $(-3, 3\sqrt{3})$ to polar coordinates (r, θ) with $r > 0$ and $0^\circ \leq \theta < 360^\circ$.
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Multiple choice — easy

- 10 If $\sin x = \frac{12}{37}$ and x is in Quadrant I (so $\cos x = \frac{35}{37}$), find $\sin 2x$.
- a. $\frac{840}{1369}$
 - b. $\frac{1081}{1369}$
 - c. $-\frac{1081}{1369}$
 - d. $-\frac{840}{1369}$
- 11 If $\sin x = \frac{11}{61}$ and x is in Quadrant I (so $\cos x = \frac{60}{61}$), find $\sin 2x$.
- a. $-\frac{1320}{3721}$
 - b. $\frac{3479}{3721}$
 - c. $\frac{1320}{3721}$
 - d. $-\frac{3479}{3721}$
- 12 If $\sin x = \frac{7}{25}$ and x is in Quadrant I (so $\cos x = \frac{24}{25}$), find $\sin 2x$.
- a. $-\frac{527}{625}$
 - b. $\frac{527}{625}$
 - c. $-\frac{336}{625}$
 - d. $\frac{336}{625}$
- 13 If $\sin x = \frac{8}{17}$ and x is in Quadrant I (so $\cos x = \frac{15}{17}$), find $\sin 2x$.
- a. $\frac{161}{289}$
 - b. $\frac{240}{289}$
 - c. $-\frac{161}{289}$
 - d. $-\frac{240}{289}$

- 14 If $\sin x = \frac{15}{17}$ and x is in Quadrant I (so $\cos x = \frac{8}{17}$), find $\sin 2x$.
- a. $\frac{161}{289}$
 - b. $-\frac{161}{289}$
 - c. $-\frac{240}{289}$
 - d. $\frac{240}{289}$
- 15 If $\sin x = \frac{3}{5}$ and x is in Quadrant I (so $\cos x = \frac{4}{5}$), find $\sin 2x$.
- a. $-\frac{7}{25}$
 - b. $-\frac{24}{25}$
 - c. $\frac{7}{25}$
 - d. $\frac{24}{25}$
- 16 If $\sin x = \frac{5}{13}$ and x is in Quadrant I (so $\cos x = \frac{12}{13}$), find $\sin 2x$.
- a. $-\frac{120}{169}$
 - b. $-\frac{119}{169}$
 - c. $\frac{119}{169}$
 - d. $\frac{120}{169}$
- 17 If $\sin x = \frac{4}{5}$ and x is in Quadrant I (so $\cos x = \frac{3}{5}$), find $\sin 2x$.
- a. $\frac{24}{25}$
 - b. $\frac{7}{25}$
 - c. $-\frac{7}{25}$
 - d. $-\frac{24}{25}$

- 18 If $\sin x = \frac{9}{41}$ and x is in Quadrant I (so $\cos x = \frac{40}{41}$), find $\sin 2x$.
- a. $\frac{720}{1681}$
 - b. $-\frac{1519}{1681}$
 - c. $-\frac{720}{1681}$
 - d. $\frac{1519}{1681}$
- 19 If $\sin x = \frac{60}{61}$ and x is in Quadrant I (so $\cos x = \frac{11}{61}$), find $\sin 2x$.
- a. $\frac{3479}{3721}$
 - b. $-\frac{3479}{3721}$
 - c. $\frac{1320}{3721}$
 - d. $-\frac{1320}{3721}$
- 20 If $\sin x = \frac{20}{29}$ and x is in Quadrant I (so $\cos x = \frac{21}{29}$), find $\sin 2x$.
- a. $\frac{41}{841}$
 - b. $\frac{840}{841}$
 - c. $-\frac{840}{841}$
 - d. $-\frac{41}{841}$
- 21 If $\sin x = \frac{21}{29}$ and x is in Quadrant I (so $\cos x = \frac{20}{29}$), find $\sin 2x$.
- a. $-\frac{840}{841}$
 - b. $\frac{840}{841}$
 - c. $\frac{41}{841}$
 - d. $-\frac{41}{841}$

- 22 If $\sin x = \frac{40}{41}$ and x is in Quadrant I (so $\cos x = \frac{9}{41}$), find $\sin 2x$.
- a. $-\frac{1519}{1681}$
 - b. $\frac{1519}{1681}$
 - c. $-\frac{720}{1681}$
 - d. $\frac{720}{1681}$
- 23 If $\sin x = \frac{24}{25}$ and x is in Quadrant I (so $\cos x = \frac{7}{25}$), find $\sin 2x$.
- a. $-\frac{336}{625}$
 - b. $\frac{527}{625}$
 - c. $-\frac{527}{625}$
 - d. $\frac{336}{625}$
- 24 If $\sin x = \frac{12}{13}$ and x is in Quadrant I (so $\cos x = \frac{5}{13}$), find $\sin 2x$.
- a. $-\frac{120}{169}$
 - b. $\frac{120}{169}$
 - c. $-\frac{119}{169}$
 - d. $\frac{119}{169}$

Multiple choice — medium

- 25 A triangle has $a = 14$, $b = 14$, and included angle $C = 60^\circ$. Find c .
- a. 15
 - b. 13
 - c. 14
 - d. 20

- 26 A triangle has $a = 12$, $b = 12$, and included angle $C = 60^\circ$. Find c .
- a. 17
 - b. 13
 - c. 11
 - d. 12
- 27 A triangle has $a = 14$, $b = 16$, and included angle $C = 120^\circ$. Find c .
- a. 21
 - b. 25
 - c. 26
 - d. 27
- 28 A triangle has $a = 3$, $b = 5$, and included angle $C = 120^\circ$. Find c .
- a. 9
 - b. 8
 - c. 6
 - d. 7
- 29 A triangle has $a = 17$, $b = 17$, and included angle $C = 60^\circ$. Find c .
- a. 16
 - b. 18
 - c. 24
 - d. 17

- 30 A triangle has $a = 13$, $b = 13$, and included angle $C = 60^\circ$. Find c .
- a. 12
 - b. 14
 - c. 18
 - d. 13
- 31 A triangle has $a = 5$, $b = 12$, and included angle $C = 90^\circ$. Find c .
- a. 13
 - b. 12
 - c. 15
 - d. 14
- 32 A triangle has $a = 9$, $b = 9$, and included angle $C = 60^\circ$. Find c .
- a. 13
 - b. 8
 - c. 10
 - d. 9
- 33 A triangle has $a = 16$, $b = 16$, and included angle $C = 60^\circ$. Find c .
- a. 16
 - b. 23
 - c. 15
 - d. 17

- 34 A triangle has $a = 6$, $b = 8$, and included angle $C = 90^\circ$. Find c .
- a. 11
 - b. 10
 - c. 9
 - d. 12
- 35 A triangle has $a = 6$, $b = 6$, and included angle $C = 60^\circ$. Find c .
- a. 5
 - b. 6
 - c. 8
 - d. 7
- 36 A triangle has $a = 9$, $b = 15$, and included angle $C = 120^\circ$. Find c .
- a. 17
 - b. 21
 - c. 20
 - d. 22
- 37 A triangle has $a = 5$, $b = 5$, and included angle $C = 60^\circ$. Find c .
- a. 4
 - b. 6
 - c. 7
 - d. 5

38 A triangle has $a = 10$, $b = 10$, and included angle $C = 60^\circ$. Find c .

- a. 14
- b. 10
- c. 11
- d. 9

39 A triangle has $a = 5$, $b = 8$, and included angle $C = 60^\circ$. Find c .

- a. 8
- b. 6
- c. 9
- d. 7

Multiple choice — hard

40 In a triangle, $\angle A = 135^\circ$, $a = 16$, and $\angle B = 45^\circ$. Find b .

- a. 17
- b. 32
- c. 16
- d. 18

41 In a triangle, $\angle A = 90^\circ$, $a = 8$, and $\angle B = 30^\circ$. Find b .

- a. 4
- b. 6
- c. 8
- d. 5

- 42 In a triangle, $\angle A = 135^\circ$, $a = 3$, and $\angle B = 45^\circ$. Find b .
- a. 3
 - b. 6
 - c. 4
 - d. 5
- 43 In a triangle, $\angle A = 135^\circ$, $a = 17$, and $\angle B = 45^\circ$. Find b .
- a. 19
 - b. 18
 - c. 34
 - d. 17
- 44 In a triangle, $\angle A = 30^\circ$, $a = 12$, and $\angle B = 90^\circ$. Find b .
- a. 12
 - b. 48
 - c. 24
 - d. 25
- 45 Convert the rectangular point $(7, 24)$ to polar coordinates (r, θ) with $r > 0$. Find r .
- a. 24
 - b. 27
 - c. 7
 - d. 25

- 46 Convert the rectangular point $(3, -4)$ to polar coordinates (r, θ) with $r > 0$. Find r .
- a. 7
 - b. 3
 - c. 5
 - d. 4
- 47 Convert the rectangular point $(12, 35)$ to polar coordinates (r, θ) with $r > 0$. Find r .
- a. 37
 - b. 39
 - c. 35
 - d. 12
- 48 Convert the rectangular point $(-8, -15)$ to polar coordinates (r, θ) with $r > 0$. Find r .
- a. 17
 - b. 15
 - c. 8
 - d. 19
- 49 Convert the rectangular point $(12, -35)$ to polar coordinates (r, θ) with $r > 0$. Find r .
- a. 12
 - b. 37
 - c. 35
 - d. 39

50 If $\sin x = \frac{12}{37}$, find $\cos 2x$.

a. $\frac{840}{1369}$

b. $-\frac{840}{1369}$

c. $-\frac{1081}{1369}$

d. $\frac{1081}{1369}$

51 If $\sin x = \frac{8}{17}$, find $\cos 2x$.

a. $\frac{161}{289}$

b. $\frac{240}{289}$

c. $-\frac{240}{289}$

d. $-\frac{161}{289}$

52 If $\sin x = \frac{7}{25}$, find $\cos 2x$.

a. $-\frac{527}{625}$

b. $-\frac{336}{625}$

c. $\frac{527}{625}$

d. $\frac{336}{625}$

53 If $\sin x = \frac{35}{37}$, find $\cos 2x$.

a. $\frac{840}{1369}$

b. $\frac{1081}{1369}$

c. $-\frac{1081}{1369}$

d. $-\frac{840}{1369}$

54 If $\sin x = \frac{15}{17}$, find $\cos 2x$.

a. $-\frac{240}{289}$

b. $\frac{161}{289}$

c. $-\frac{161}{289}$

d. $\frac{240}{289}$